Free multi-domain diffusion models and the thermal history of Win-06-01b goethite from the water-saturated step heat HT-Win-06-01b

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Abstract

We invert the thermal history of Win-06-01b goethite from the water-saturated step-heating experiment HT-Win-06-01b in two stages. First, multi-domain diffusion (MDD) models of $N = 1$–10 spherical domains, with all ^{3}He fractions and sizes free, are fitted to the ^{3}He release alone (Farley and Flowers, 2012). The domain structure stabilizes at $N = 4$: 97% of the ^{3}He resides in one domain slightly larger than the Stage 1 reference domain ($r/r_0 = 1.12$), and three small domains holding 2.8% account for the high early diffusivities. Second, with that structure fixed, thermal histories are fitted to the bulk-normalized ^{4}He/^{3}He spectrum and checked against the (U–Th)/He age of 130–180 Ma. Steady linear cooling to 25 °C fits best ($\chi^2_\nu = 2.0$ for paths consistent with the He age): crystallization at 150–175 Ma at 76–80 °C. A burial-and-exhumation path with a Late Cretaceous peak of 58–65 °C fits less well ($\chi^2_\nu \approx 4.8$–5.1) but agrees with regional apatite fission-track constraints. An isothermal hold fits poorly ($\chi^2_\nu \approx 18$). The cooling solution favours crystallization from warm (hydrothermal) fluids over formation at the surface.

1 Motivation and data

Win-06-01b is dense colloform goethite collected as float in the Flinders Ranges, South Australia, with a (U–Th)/He age of 130–180 Ma (Waltenberg, 2012). Apatite fission-track studies in the region record cooling from roughly 50–70 °C after the mid-Cretaceous (Gibson and Stüwe, 2000; Mitchell et al., 1998), and Triassic intramontane basins less than 1.5 km thick indicate that parts of the ranges were near the surface in the Early Mesozoic (Mitchell et al., 1998). A proton-irradiated aliquot was step-heated in a sealed silica ampoule under water-saturated conditions (HT-Win-06-01b; 34 steps, 170–250 °C), releasing 14.8% of its ^{3}He before final fusion. The Stage 1 regression of that experiment defines the reference Arrhenius line used here ($E_a = 157.1$ kJ mol^{-1}, $\ln(D_0/a_0^2) = 16.16$ with D_0/a_0^2 in s^{-1}).

Earlier versions of this analysis split the sample into a high-retentivity domain pinned to the Stage 1 line and a low-retentivity assemblage holding at most 7.5% of the ^{3}He. Here that distinction is dropped: all domains are equivalent, and each has a free ^{3}He fraction and size.

2 Methods

2.1 Domain model

The sample is represented by N non-interacting spherical domains that share the Stage 1 activation energy and frequency factor. Domain j holds a fraction f_j of the uniformly distributed, proton-

induced ^{3}He ($\sum_j f_j = 1$) and has radius r_j relative to the reference radius r_0 implied by the Stage 1 regression, so that

$$\left(\frac{D}{a^2}\right)_j = \left(\frac{r_j}{r_0}\right)^{-2} \frac{D_0}{a_0^2} \exp\left(-\frac{E_a}{RT}\right). \tag{1}$$

A model with N domains has $k = 2N-1$ free parameters. Sizes are bounded by $10^{-6} \le r_j/r_0 \le 10^3$.

2.2 Laboratory release of ^{3}He

For a heating schedule of steps i with temperatures T_i and durations Δt_i, the reference domain accumulates the non-dimensional time

$$\tau_{0,i} = \sum_{m \le i} \frac{D_0}{a_0^2} \exp\left(-\frac{E_a}{RT_m}\right) \Delta t_m, \tag{2}$$

and domain j reaches $\tau_{0,i}(r_j/r_0)^{-2}$. An initially uniform sphere releases the cumulative fraction (Crank, 1979)

$$F(\tau) = 1 - \frac{6}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{k^2} \exp\left(-k^2\pi^2\tau\right), \tag{3}$$

so the model cumulative and fractional ^{3}He yields of the sample are

$$\Sigma F_i = \sum_{j=1}^{N} f_j \, F\left(\tau_{0,i}\,(r_j/r_0)^{-2}\right), \qquad \Delta F_i = \Sigma F_i - \Sigma F_{i-1}. \tag{4}$$

2.3 Fitting the domain structure to ^{3}He

Following Farley and Flowers (2012, supplement A2), the fractional and cumulative yields are weighted equally, and $\{f_j, r_j\}$ minimize

$$S = \sum_{i=1}^{34} \left(\Delta F_i^{\mathrm{mod}} - \Delta F_i^{\mathrm{obs}}\right)^2 + \sum_{i=1}^{34} \left(\Sigma F_i^{\mathrm{mod}} - \Sigma F_i^{\mathrm{obs}}\right)^2, \tag{5}$$

with yields expressed as fractions of the total ^{3}He, including the fusion step. The six steps with less than 0.1 pccSTP of ^{3}He, which are excluded throughout the chapter, are left out of both sums (28 steps remain). Their gas still counts toward the cumulative yields of later steps, and their durations toward the laboratory schedule. Models are built up one domain at a time: the N-domain search starts from the $(N-1)$-domain solution plus one new domain trialled at 15 sizes and two fractions, together with ten random starts. The fit therefore cannot get worse as N increases. We add domains until a new domain no longer changes the fit or the domain structure. The Arrhenius and delta plots are not fitted. They follow from applying the single-sphere inversion of Fechtig and Kalbitzer (1966) to the model cumulative release. Their misfit is reported as a reduced χ^2 against the bounding-method uncertainties in $\ln(D/a^2)$ over the same 28 steps.

2.4 Radiogenic ^{4}He in nature

Each domain produces ^{4}He uniformly at rate P and loses it by diffusion with a zero-concentration boundary,

$$\frac{\partial C}{\partial t} = \frac{D}{a^2} \frac{1}{x^2} \frac{\partial}{\partial x}\left(x^2 \frac{\partial C}{\partial x}\right) + P, \qquad C(1,t) = 0, \quad C(x, t_{\mathrm{form}}) = 0, \tag{6}$$

where x is radial position normalized to the domain radius and $t_{\rm form}$ is the crystallization age. With $C = \sum_k A_k(t)\sin(k\pi x)/x$ (Carslaw and Jaeger, 1959), each mode evolves as

$$\frac{\mathrm{d}A_k}{\mathrm{d}t} = -k^2\pi^2\frac{D}{a^2}A_k + c_k P, \qquad c_k = \frac{2(-1)^{k+1}}{k\pi}, \tag{7}$$

which over an interval Δt (2 Myr) at constant temperature has the exact solution

$$A_k \leftarrow A_k e^{-\lambda_k} + c_k P\,\Delta t\,\frac{1-e^{-\lambda_k}}{\lambda_k}, \qquad \lambda_k = k^2\pi^2\frac{D}{a^2}\Delta t. \tag{8}$$

We use 8000 modes. The ^{4}He content per unit volume of domain j is $N_j = 3\sum_k A_{k,j}(-1)^{k+1}/(k\pi)$. Taking U and Th to be distributed like the ^{3}He and neglecting parent decay ($\sim$3% over 200 Myr), the bulk He age is

$$t_{\rm He} = \frac{1}{P}\sum_j f_j\,N_j. \tag{9}$$

2.5 Laboratory degassing of the natural profiles

In the laboratory each mode decays as $\exp(-k^2\pi^2\tau)$. The reference times $\tau_{0,i}$ used for the ^{4}He prediction are solved from the *measured* cumulative ^{3}He by inverting Eq. 4. The predicted spectrum is therefore independent of the heating schedule and of step-to-step kinetic scatter:

$$\frac{R_{{\rm step},i}}{R_{\rm bulk}} = \frac{\Delta N_{4,i}/N_{4,\rm tot}}{\Delta F_i^{\rm obs}}, \qquad \Delta N_{4,i} = \sum_j f_j\left[N_j(\tau_{0,i-1}) - N_j(\tau_{0,i})\right], \tag{10}$$

where $N_j(\tau)$ is the ^{4}He remaining in domain j after laboratory time $\tau(r_j/r_0)^{-2}$ and $N_{4,\rm tot} = \sum_j f_j N_j(0)$ includes the gas released at fusion, consistent with the measured bulk ratio. Because the eigenfunctions do not depend on D, the present-day amplitudes $A_{k,j}$ carry each domain's non-uniform ^{4}He profile into the laboratory unchanged. No assumption of a uniform ^{4}He distribution is made. An independent finite-volume solution reproduces the predicted spectrum to 8×10^{-4} in $R/R_{\rm bulk}$ (Fig. 6).

2.6 Thermal histories

The thermal history is inverted with the domain structure fixed, so it must work within the bounds set by the ^{3}He inversion. We compare three families of piecewise-linear thermal history, each with two free parameters.

- **Steady cooling:** T falls linearly from T_0 at $t_{\rm form}$ to 25 °C today. The goethite forms at an elevated temperature, for example from warm fluids at depth, and cools monotonically. Free parameters: $t_{\rm form}$ and T_0.
- **Burial and exhumation:** through $(t_{\rm form}, T_s)$, $(t_{\rm peak}, T_{\rm max})$, (50 Ma, T_s) and $(0, T_s)$ with $T_s = 20$ °C. The goethite forms at the surface, is buried and heated to a peak temperature, and is exhumed by the Paleogene, when Paleogene sediments in Flinders Ranges valleys show that basement was exposed (Mitchell et al., 1998). Free parameters: $t_{\rm peak}$ and $T_{\rm max}$, for fixed crystallization ages of 180, 195 and 210 Ma.
- **Isothermal hold:** $T = T_{\rm iso}$ from $t_{\rm form}$ to the present. Free parameters: $t_{\rm form}$ and $T_{\rm iso}$.

Each path is scored against

$$\chi^2 = \sum_i \left(\frac{R_i^{\mathrm{mod}} - R_i^{\mathrm{M}}}{\sigma_i} \right)^2, \qquad \sigma_i = \max\left[\tfrac{1}{2} \left(R_i^{\mathrm{UB}} - R_i^{\mathrm{LB}} \right), 0.02 \right], \tag{11}$$

over the 24 steps that have lag-correction bounds and more than 0.1 pccSTP of ^{3}He, so $\nu = 22$. M, LB and UB are the central, lower-bound and upper-bound bulk-normalized ratios of the bounding method. χ^2 is mapped on regular grids:

- burial: t_{peak} from 55 Ma to $t_{\mathrm{form}} - 2.5$ Ma in 2.5 Myr steps, and $T_{\mathrm{max}} = 30$–$90\,^{\circ}\mathrm{C}$ in $1\,^{\circ}\mathrm{C}$ steps;
- steady cooling and isothermal hold: $t_{\mathrm{form}} = 50$–500 Ma in 5 Myr steps, and $T_0 = 25$–$120\,^{\circ}\mathrm{C}$ or $T_{\mathrm{iso}} = 0$–$70\,^{\circ}\mathrm{C}$ in $1\,^{\circ}\mathrm{C}$ steps.

Natural histories are integrated in 1–2 Myr intervals of constant temperature, and the predicted He age is contoured at the limits of the measured range. The joint 1σ region of each map is $\chi^2 \leq \chi^2_{\mathrm{min}} + 2.30 \max(1, \chi^2_\nu)$. We treat the He age as a hard constraint. The best fit of each family is the lowest-χ^2 path with a predicted age of 130–180 Ma, and parameter ranges are those of the age-consistent paths within the same $\Delta\chi^2$ of that best fit.

3 Results

3.1 Domain structure from ^{3}He

Table 1 lists the fits for all N, and Fig. 1 shows the fitted domain spectra.

- $N = 1$. A single domain cannot fit the release (Arrhenius $\chi^2_\nu = 27$).
- $N = 2$. The second domain captures the high early diffusivities but leaves a poor Arrhenius fit ($\chi^2_\nu = 3.3$).
- $N = 3$. The yield misfit falls by more than half and the Arrhenius χ^2_ν falls to 1.08.
- $N = 4$. The smallest domain splits in two. The fit improves slightly further (Arrhenius $\chi^2_\nu = 1.00$).
- $N \geq 5$. Every added domain coincides in size with the dominant domain. The fit is unchanged.

We therefore adopt $N = 4$ as the complete description of the ^{3}He data. Its domains are:

- 0.57% of the ^{3}He at $r/r_0 = 6.8 \times 10^{-4}$;
- 0.66% at 2.9×10^{-3};
- 1.5% at 0.031;
- 97.2% at 1.12.

The three small domains are fully degassed during the experiment. The dominant domain releases 12% of its ^{3}He, at a diffusivity 0.80 times the Stage 1 line. The Stage 1 line is therefore a slightly fast description of the dominant domain: it was regressed on bulk-inverted diffusivities that still carry a small contribution from the fast domains. The model reproduces the Arrhenius array, including the retrograde segments, and the gradual approach of the delta plot to the Stage 1 line (Fig. 2). Figure 5 shows the fitted yields and the resulting Arrhenius and delta plots for $N = 1$–5.

Table 1: Multi-domain fits to the ^{3}He release (N domains, $k = 2N - 1$ parameters). RMS is the yield misfit ($\times 10^{-4}$, fraction of total ^{3}He) computed over the 28 steps above the signal threshold. "Fast" is the ^{3}He fraction in domains with $r/r_0 < 0.1$; f_{main} and r_{main}/r_0 are the fraction and size of the largest-fraction domain. For $N \geq 5$, the dominant domain is split among several domains of identical size.

N	k	RMS	Arrh. χ^2_ν	fast (%)	f_{main} (%)	r_{main}/r_0
1	1	85.30	27.45	0.0	100.0	0.80
2	3	20.20	3.30	2.2	97.8	1.05
3	5	8.45	1.08	2.7	97.3	1.12
4	7	7.81	1.00	2.8	97.2	1.12
5	9	7.81	1.11	2.8	95.1	1.12
6	11	7.81	1.24	2.8	85.6	1.12
7	13	7.81	1.40	2.8	85.3	1.12
8	15	7.81	1.62	2.8	76.8	1.12
9	17	7.81	1.91	2.8	76.5	1.12
10	19	7.81	2.34	2.8	74.6	1.12

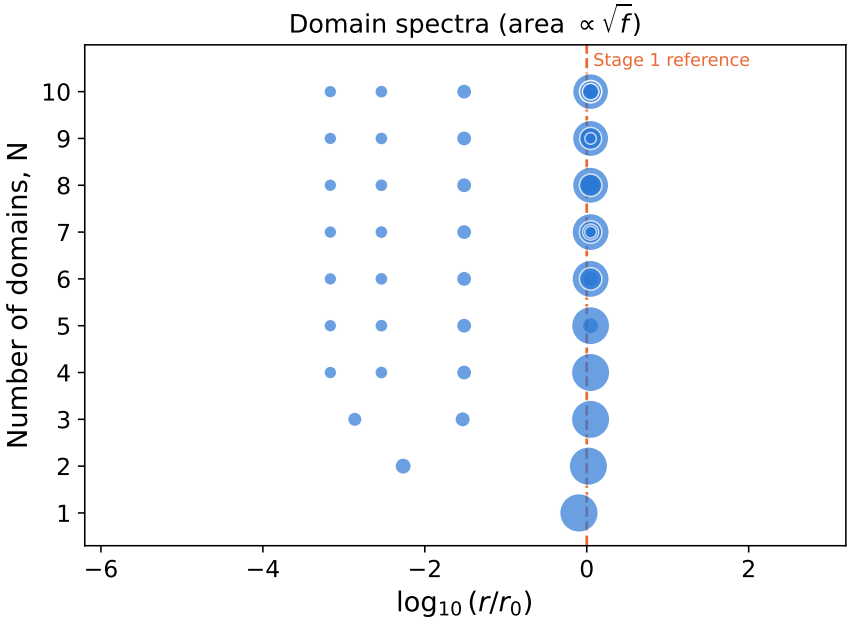


Figure 1: Domain spectra of the best-fitting models. Symbol area scales with the square root of the ^{3}He fraction; the dashed line marks the Stage 1 reference domain. The three small domains are unchanged from $N = 4$ onward; domains added beyond $N = 4$ coincide in size with the dominant domain.

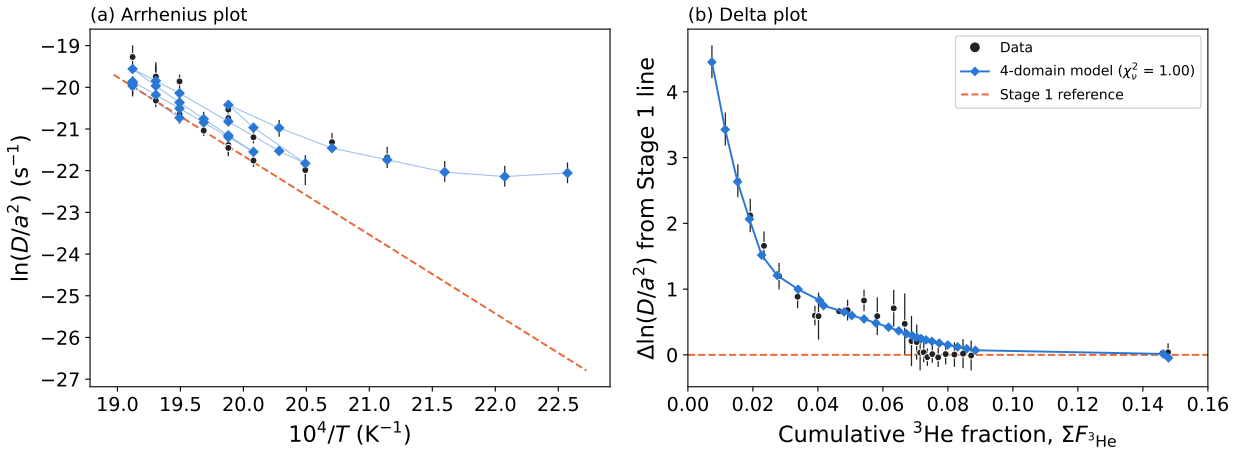


Figure 2: (a) Arrhenius plot and (b) delta plot (departure from the Stage 1 line) of the measured ^{3}He diffusivities, with the values implied by the four-domain fit to the ^{3}He yields. Neither plot is fitted directly. The six low-signal steps are omitted.

3.2 Thermal history from ^{4}He/^{3}He

Table 2 gives the best fit of each family with the four-domain structure fixed. Figure 3 compares the best steady-cooling and burial predictions with the spectrum, and Fig. 4 shows their χ^2 maps on a common colour scale.

Table 2: Thermal histories fitted to the ^{4}He/^{3}He spectrum with the four-domain structure fixed. Each row gives the best path with a predicted He age of 130–180 Ma ($\nu = 22$). For steady cooling, $T_{\max} = T_0$ is reached at crystallization, so $t_{\text{peak}} = t_{\text{form}}$. For the isothermal hold, $T_{\max} = T_{\text{iso}}$ throughout. Burial crystallization ages are fixed. The last three columns give the ranges of the age-consistent paths within the joint 1σ region about the best fit.

Path	t_{form} (Ma)	t_{peak} (Ma)	$T_{\max}$ (°C)	χ^2_ν	t_{He} (Ma)	t_{form} (1σ)	t_{peak} (1σ)	$T_{\max}$ (1σ)
Linear cooling	150	150	78	2.00	132	150–175	150–175	76–80
Burial	180	70.0	61	5.11	164	fixed	60–82	58–65
Burial	195	80.0	61	4.81	177	fixed	68–98	58–65
Burial	210	115.0	71	6.23	180	fixed	105–130	70–74
Isothermal hold	150	–	41	18.5	141	140–190	–	39–43

Steady cooling. Linear cooling to 25 °C fits the spectrum best. The best age-consistent path starts at 78 °C at 150 Ma ($\chi^2_\nu = 2.00$), and the age-consistent 1σ range is crystallization at 150–175 Ma at 76–80 °C, cooling at ∼0.3 °C Myr^{-1}. This path passes through ∼55 °C at 90 Ma, inside the apatite fission-track window. The spectrum alone prefers somewhat younger, hotter paths: the unconstrained minimum ($\chi^2_\nu = 1.68$; 89 °C at 80 Ma) predicts a He age of 68 Ma, and the unconstrained 1σ region lies entirely at ages younger than 130 Ma (Fig. 4b). Requiring the measured age costs $\Delta\chi^2 \approx 7$.

Burial and exhumation. Burial paths fit the spectrum less well ($\chi^2_\nu = 4.8$–6.2), but they agree with independent regional constraints. For crystallization at 180 and 195 Ma, the best age-consistent paths peak at 61 °C at 70 and 80 Ma, with 1σ ranges of 58–65 °C between 98 and 60 Ma

(Fig. 4a). These peaks fall in or just after the apatite fission-track window and within the 50–70 °C range of Gibson and Stüwe (2000), at the <60 °C limit reported near Copley (Mitchell et al., 1998). For crystallization at 210 Ma, only hotter and earlier peaks (70–74 °C at 105–130 Ma) give an age within the measured range, and they fit worse ($\chi^2_\nu = 6.2$).

Isothermal hold. The spectrum constrains only the product of diffusivity and time, so the χ^2 minimum is a ridge from ∼47 °C at 50 Ma to ∼35 °C at 500 Ma along which $\chi^2_\nu \approx 18.5$ throughout. No isothermal path reproduces the shape of the spectrum: they underpredict the ratios of the early steps and overpredict those of the late steps. We do not consider this family further.

Where the families differ. The steady-cooling and burial predictions differ mainly in the earliest steps ($\Sigma F_{^3\mathrm{He}} \lesssim 0.035$; Fig. 3). These steps sample the outer rim of the dominant domain. On the burial path, the sample sits at 20 °C for the last 50 Myr, long enough for radiogenic ^{4}He to reaccumulate in the rim, so the early ratios are overpredicted. Steady cooling keeps the rim depleted until geologically recent times, with diffusive loss declining monotonically toward the present. Elsewhere both leave the same residual pattern: ratios underpredicted between $\Sigma F_{^3\mathrm{He}} = 0.04$ and 0.06, and overpredicted in the two longest high-temperature steps.

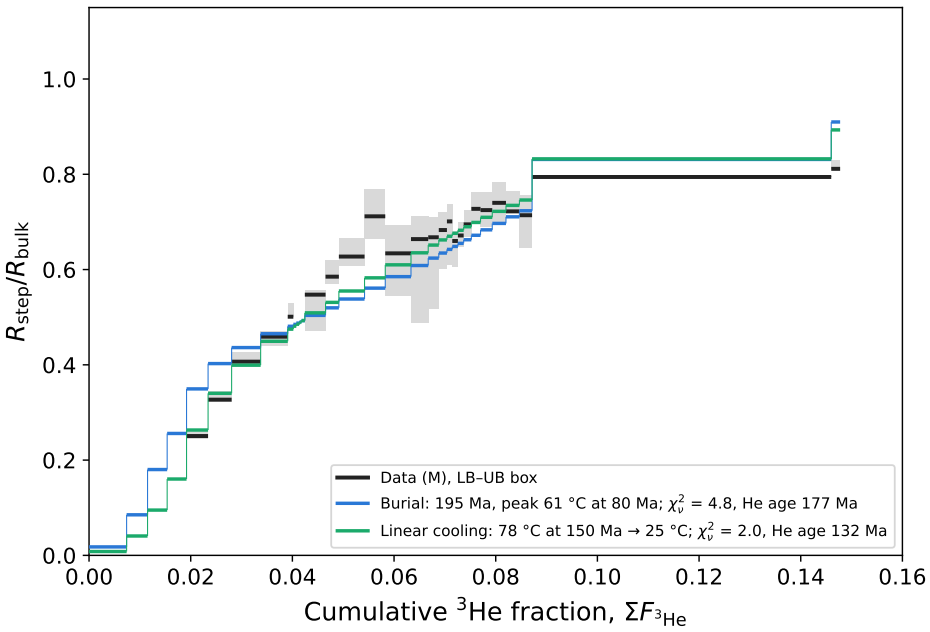


Figure 3: Bulk-normalized ^{4}He/^{3}He spectrum (black, bounding-method LB–UB boxes) with the best age-consistent burial path for crystallization at 195 Ma (blue) and the best age-consistent steady-cooling path (green), for the four-domain structure.

4 Discussion

Robust conclusions.

- The ^{3}He release is described completely by four domains. The dominant domain holds ≈97% of the ^{3}He and degasses slightly more slowly than the Stage 1 line. Fast domains hold only

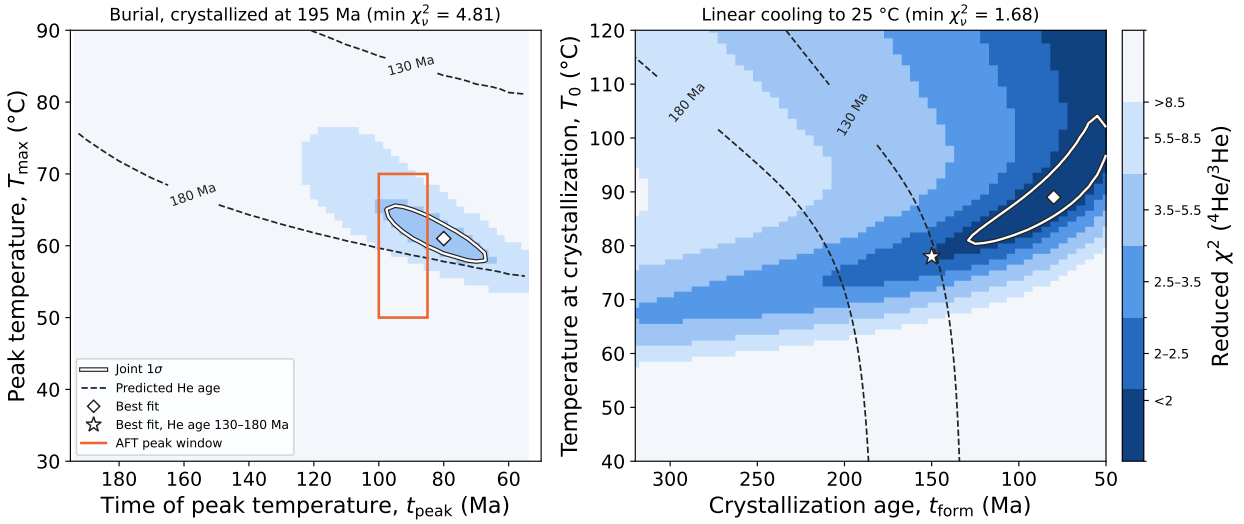


Figure 4: Reduced χ^2 of the ^{4}He/^{3}He spectrum for (a) burial paths with crystallization at 195 Ma, as a function of the time and magnitude of peak temperature, and (b) steady cooling to 25 °C, as a function of crystallization age and initial temperature. Both panels share one colour scale, anchored to the overall minimum. White outlines show the joint 1σ region of each map, dashed lines the predicted bulk He age at 130 and 180 Ma, diamonds the best fit, and the star the best fit with a He age of 130–180 Ma. The orange box in (a) is the apatite fission-track peak window (Gibson and Stüwe, 2000; Mitchell et al., 1998).

2.8%.

- Within that structure, the spectrum requires a ^{4}He-depleted rim in the dominant domain. The rim reflects diffusive loss at temperatures well above the surface during the Mesozoic, followed by loss that declined toward the present rather than stopping early.
- Both the steady-cooling and the burial solutions place the sample at ∼55–65 °C in the Late Cretaceous, consistent with the apatite fission-track record of 50–70 °C before cooling after ∼85 Ma (Gibson and Stüwe, 2000).
- Both require crystallization in the Jurassic or earliest Cretaceous: at 150–175 Ma for steady cooling, and at or after ∼195 Ma for burial.

Formation setting. The two families encode different formation settings. Burial assumes the goethite formed at the surface, for example in a weathering profile, and was later buried. Steady cooling has it forming at ∼76–80 °C. The better fit of steady cooling therefore favours crystallization from warm fluids, in a hydrothermal system or at depth, over formation at the surface. Burial nevertheless remains a viable alternative: it is the setting assumed by most goethite (U–Th)/He studies, and it matches the independent fission-track constraints closely. Two further qualifications apply.

- A hydrothermal system cools to ambient temperature in far less than a million years, so the 150 Myr linear decline is not the cooling of the system itself. It implies that the sample stayed in a warm subsurface setting after crystallization and cooled slowly, by exhumation or a declining geotherm. The two-parameter family cannot separate the fluid temperature at crystallization from the ambient temperature shortly afterwards.

- The comparison is between parameterized families. A burial path with a warmer Cenozoic, or a brief hydrothermal pulse followed by slow cooling, has not been tested and might fit as well.

Remaining misfit. Even the steady-cooling solution leaves systematic residuals. The intermediate domain ($r/r_0 \approx 0.03$, 1.5% of the ^{3}He) degasses at $\Sigma F_{^3\mathrm{He}} \approx 0.02$–$0.06$. With the common activation energy, it would have lost most of its radiogenic ^{4}He in nature, which depresses the predicted ratios where the measured ratios are highest. Diffusion-only remedies that preserve the ^{3}He fit are:

- a higher activation energy for the intermediate domain, which the retrograde segments of the experiment could test;
- domain-dependent ^{4}He production, for example from U–Th zoning across the colloform bands.

Each adds one parameter. A two-domain structure fits the burial spectrum better ($\chi^2_\nu \approx 1.3$) but reproduces the ^{3}He Arrhenius array poorly (Fig. 5), so we do not adopt it.

Caveats.

- The temperature scale rests on extrapolating the Stage 1 activation energy from 200–250 °C to near-surface temperatures ($\pm \sim$4 °C near 60 °C, not propagated here).
- Surface temperatures (20 °C for burial, 25 °C for steady cooling) and the burial exhumation time (50 Ma) are fixed.
- Domains that barely degas in the experiment are constrained only through their summed fraction.
- The model takes U and Th to be distributed like the ^{3}He, both within and among domains.

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A Supplementary figures

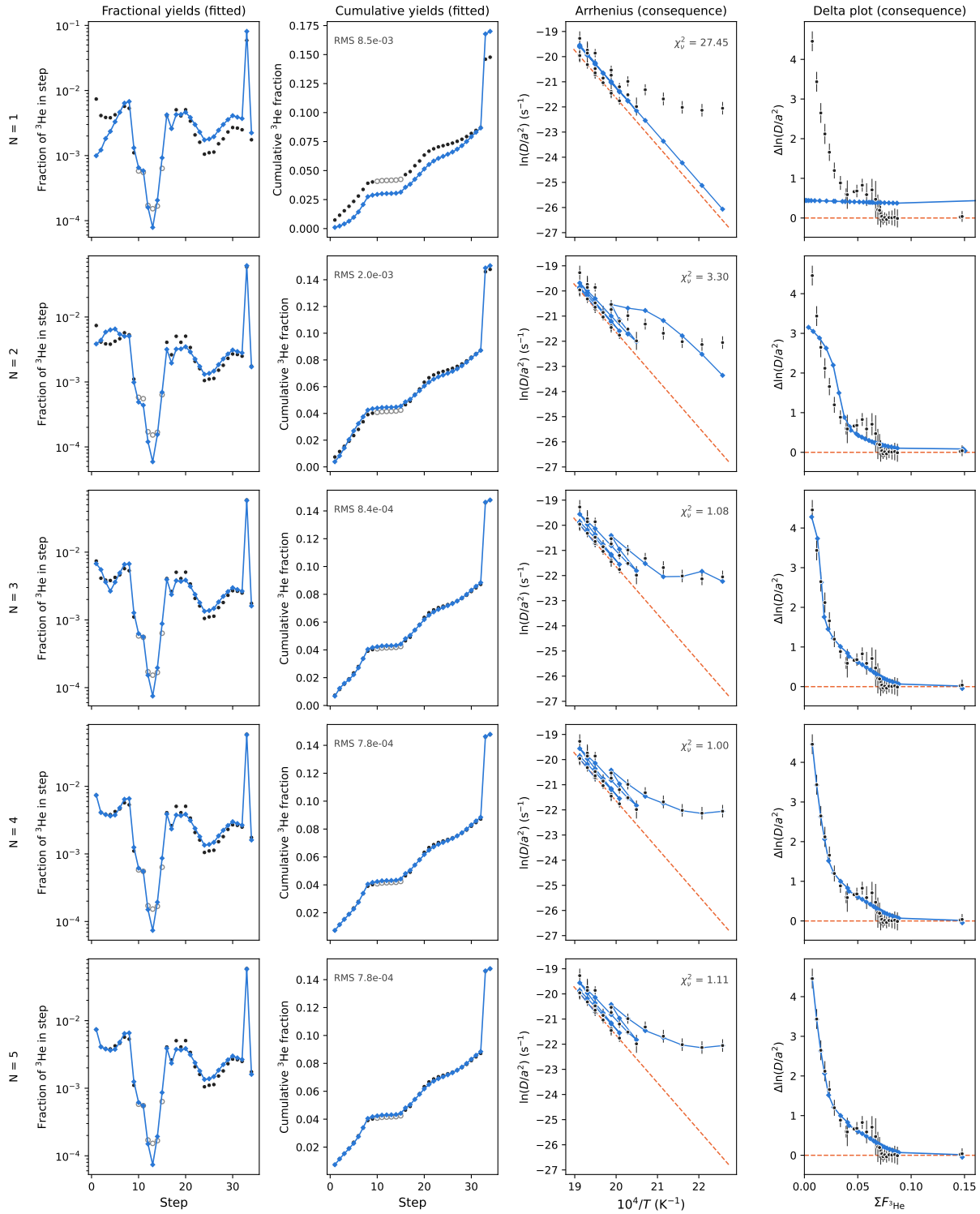


Figure 5: Fits to the ^{3}He data for $N = 1$–5 domains (rows). Columns: fitted fractional and cumulative yields, and the Arrhenius and delta plots implied by each fit. Hollow grey symbols are the six low-signal steps, which are not fitted and are omitted from the Arrhenius and delta plots.

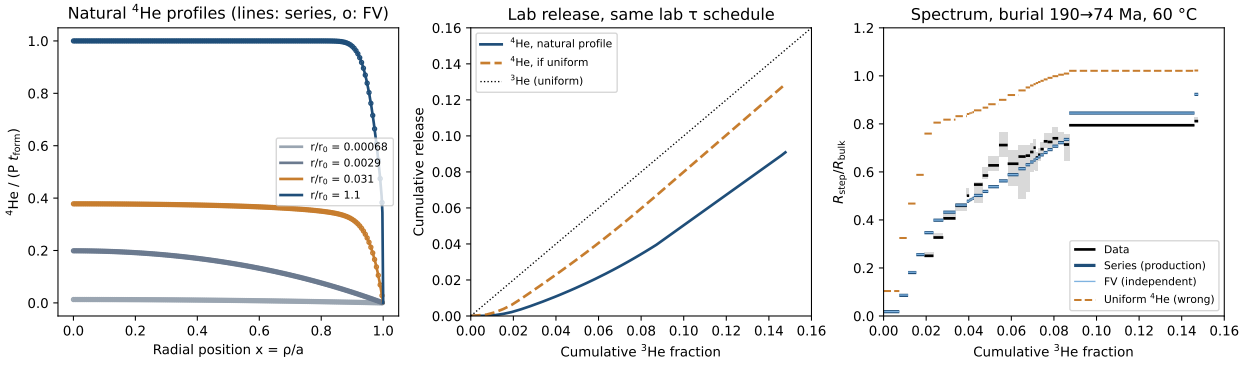


Figure 6: Check of the laboratory degassing of the natural ^{4}He profiles, for the four-domain structure on the burial path 190→74 Ma, 60 °C. Left: present-day ^{4}He profiles from the eigenfunction series (lines) and an independent finite-volume solution (2000 shells, symbols). Middle: cumulative laboratory release on the same schedule, for the natural profiles and for the same ^{4}He spread uniformly. Right: the series and finite-volume spectra agree to 8×10^{-4} in R/R_{bulk}; assuming uniform ^{4}He would raise χ^2 from 109 to ∼3500. Truncating the series at 500 modes changes the spectrum by $< 10^{-4}$.